Josephson effect — Solution

X25 (2025) — English translation

A1 **0.50 pts**

Let at the initial moment of time $t = 0$ the current through the contact be equal to zero. At this moment, voltage V was applied to the contact. Determine the dependence of the current on time $I(t)$. Express your answer using Φ_0, V, I_c, R.

The voltage sets the derivative of the phase difference at the contact

$$\dot{\varphi} = \frac{2\pi}{\Phi_0} V.$$

Since до включения напряжения current equals нулю, initial phase $\varphi(0) = 0$. (For/At включении напряжения фазnot может изmestring/threadся мгновенно, since быстрое изменение фазы for/atводит к бесконечно большому напряжению.) Then фаза имеет вид

$$\varphi = \frac{2\pi}{\Phi_0} Vt.$$

Then полный current складывается from нормального currentа $I_n = V/R$ and сверхпроводящего currentа $I_s = I_c \sin\varphi$

Answer: \textbfAnswer:

$$I = \frac{V}{R} + I_c \sin\left(\frac{2\pi}{\Phi_0} Vt\right)$$

A2 **0.50 pts**

Let a direct current $I > I_c$ be passed through the contact. Write down the differential equation for $\varphi(t)$.

Let us write the total current as the sum of the normal current and the superconducting current, expressing the voltage through the derivative of the phase, as a result we obtain

Answer: \textbfAnswer:

$$I = I_c \sin\varphi + \frac{\Phi_0}{2\pi R} \dot{\varphi}$$

A3 **0.70 pts**

Differentiate the equation with respect to time. Eliminate the phases from the equation and obtain an expression for $\dot{V}(t)^2$ in terms of V, I and the system parameters.

Differentiating the equation, we obtain

$$(I_c \cos\varphi)\dot{\varphi} + \frac{\Phi_0}{2\pi R} \ddot{\varphi} = 0.$$

Производную $\dot{\varphi}$ let us express via/through voltage

$$V(I_c \cos\varphi) + \frac{\Phi_0}{2\pi R} \dot{V} = 0,$$

а косинус фазы – via/through синус from исходного уравнения,

$$I_c \sin\varphi = I - \frac{V}{R}, \quad I_c^2 \cos^2\varphi = I_c^2 - \left(I - \frac{V}{R}\right)^2,$$

Finally we obtain

Answer: \textbfAnswer:

$$\dot{V}^2 = \left(\frac{2\pi R}{\Phi_0}\right)^2 V^2 \left(I_c^2 - \left(I - \frac{V}{R}\right)^2\right)$$

A4 — 0.30 pts

In the previous equation, go to the variable $u = 1/V$, express $\dot{u}^2$ in terms of u, I and the system parameters.

Let us express the derivative of the voltage

$$\dot{V} = -\frac{\dot{u}}{u^2},$$

substituting в equation

$$\frac{\dot{u}^2}{u^4} = \left(\frac{2\pi R}{\Phi_0}\right)^2 \frac{1}{u^2} \left(I_c^2 - \left(I - \frac{1}{Ru}\right)^2\right).$$

Домножая на u^4, we obtain

Answer: \textbfAnswer:

$$\dot{u}^2 = \left(\frac{2\pi R}{\Phi_0}\right)^2 \left(I_c^2 u^2 - \left(Iu - \frac{1}{R}\right)^2\right)$$

A5 — 0.80 pts

You should get an equation similar to the law of conservation of energy for oscillations of a harmonic oscillator. Determine the frequency of oscillations ω, the value u_0 about which the oscillations occur, and the amplitude of oscillations A.

Differentiating the equation from the previous paragraph, we obtain the vibration equation

$$\ddot{u} = -\left(\frac{2\pi R}{\Phi_0}\right)^2 \left((I^2 - I_c^2)u - \frac{I}{R}\right).$$

From coefficienta перед u we find частоту колебаний

$$\omega = \frac{2\pi R}{\Phi_0}\sqrt{I^2 - I_0^2},$$

a also положение equalsвесия

$$u_0 = \frac{I}{R(I^2 - I_c^2)}.$$

For того, чтобы найти амплитуду колебаний, we determine значения u, for/at которых $\dot{u} = 0$, we get квадратное equation

$$(I^2 - I_c^2)u^2 - \frac{2I}{R}u + \frac{1}{R^2} = 0,$$

корни которого

$$u = \frac{1}{I^2 - I_c^2}\left(\frac{I}{R} \pm \frac{I_c}{R}\right), \quad u_1 = \frac{1}{R(I + I_c)}, \quad u_2 = \frac{1}{R(I - I_c)}.$$

Тот же result можно найти, olimitandв максимальное and минимальное значения напряжения from the equation

$$I = I_c \sin\varphi + \frac{V}{R}, \quad V_{min} = R(I - I_c), \quad V_{max} = R(I + I_c).$$

Then amplitude

$$A = \frac{1}{2}(u_2 - u_1) = \frac{I_c}{R(I^2 - I_c^2)}.$$

Answer: \textbfAnswer:

$$\omega = \frac{2\pi R}{\Phi_0}\sqrt{I^2 - I_0^2}; \quad u_0 = \frac{I}{R(I^2 - I_c^2)}; \quad A = \frac{I_c}{R(I^2 - I_c^2)}.$$

A6 — 0.20 pts

Write down the dependence $V(t)$. Consider that at $t = 0$ the value $V(t)$ has a minimum.

The dependence of the quantity u on time

$$u = u_0 + A\cos\omega t,$$

where фаза olimitяется тем conditionм, что V минимальна в initial moment времени, and therefore u is maximum.

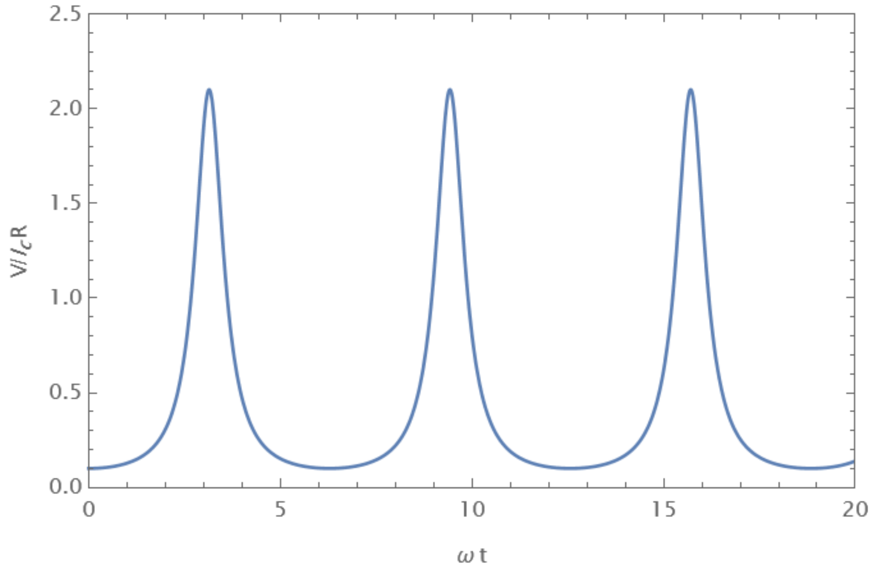
Answer: \textbfAnswer:

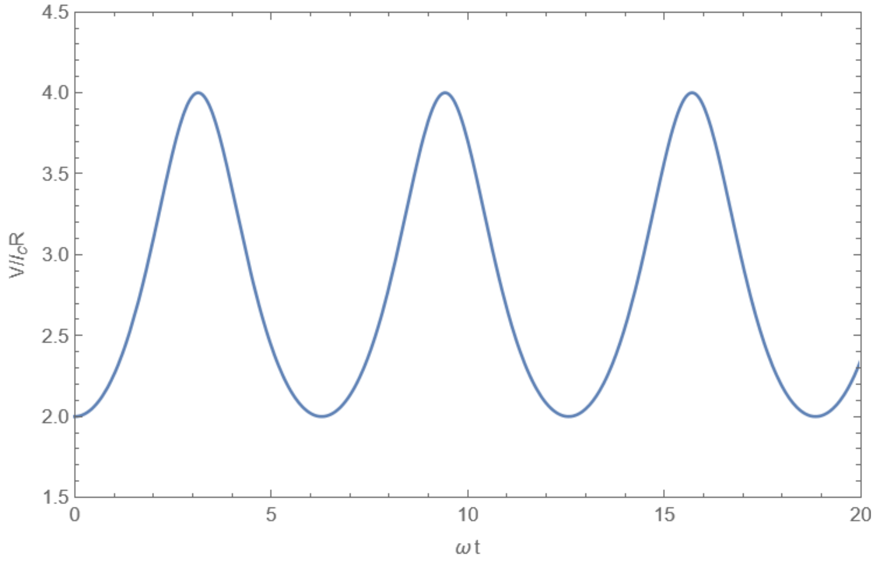
$$V = \frac{R(I^2 - I_c^2)}{I + I_c\cos\left(\frac{2\pi R}{\Phi_0}\sqrt{I^2 - I_0^2}t\right)}$$

A7 — 0.60 pts

Construct high-quality graphs of the dependence $V(t)$ for cases $I = 1.1I_c$ and $I = 3I_c$, indicate the coordinates of the characteristic points.

Answer: \textbfAnswer: Graph for case $I = 1.1I_c$





A8 **0.80 pts**

Find the average (over time) voltage across the contact. Express your answer in terms of R, I, I_c.

Let's express the voltage through the phase derivative and integrate over time

$$V = \frac{\Phi_0}{2\pi}\dot{\varphi}, \quad \int\limits_0^T V(t)dt = \frac{\Phi_0}{2\pi}(\varphi(T) - \varphi(0)).$$

If T – period колебаний, то $\varphi(T) - \varphi(0) = 2\pi$, а среднее value напряжения equals

$$\langle V \rangle = \frac{1}{T}\int\limits_0^T V(t)dt = \frac{\Phi_0}{T} = \frac{\Phi_0}{2\pi}\omega = R\sqrt{I^2 - I_0^2}.$$

Этот же result можно порayandть интегрированием явного выражения for $V(t)$

$$\langle V \rangle = \frac{1}{T}\int\limits_0^T V(t)dt = \frac{1}{T}\int\limits_0^T \frac{R(I^2 - I_c^2)}{I + I_c\cos(\omega t)}dt = \frac{1}{2\pi}\int\limits_0^{2\pi} \frac{R(I^2 - I_c^2)}{I + I_c\cos\theta}d\theta,$$

where $\theta = \omega t$. Этот интеграл можно вычислить с помощью подстановки

$$\xi = \tan\frac{\theta}{2}, \ \cos\theta = \frac{1-t^2}{1+t^2}, \ d\theta = \frac{2dt}{1+t^2}.$$

$$\int\limits_0^{2\pi} \frac{d\theta}{I + I_c\cos\theta} = \int\limits_{-\pi}^{\pi} \frac{d\theta}{I + I_c\cos\theta} = \int\limits_{-\infty}^{\infty} \frac{\frac{2dt}{1+t^2}}{I + I_c\frac{1-t^2}{1+t^2}} = 2\int\limits_{-\infty}^{\infty} \frac{dt}{I + I_c + (I - I_c)t^2} = \frac{2\pi}{\sqrt{I^2 - I_c^2}}.$$

This gives the same answer

Answer: \textbfAnswer:

$$V = R\sqrt{I^2 - I_c^2}$$

A9 — 0.70 pts

Determine the dependence of the energy $E(\varphi)$ of the Josephson junction as a function of the phase difference φ on it. The answer may also include Φ_0 and I_c. Consider that $E(0) = 0$.

Source power

$$P = IV = I_s V = I_c \sin\varphi \frac{\Phi_0}{2\pi}\dot{\varphi}.$$

Вклад нормального currentа в power имеет вид

$$P_n = I_n V = \frac{1}{R}\left(\frac{\Phi_0}{2\pi}\right)^2 \dot{\varphi}^2.$$

For/At увеличении характерного времени изменения фазы T эта power изменяется как $1/T^2$, therefore полное выделившееся тепло ведет себя как $(1/T^2)T \sim 1/T$ and его можно сделать сколь угодно малым, увеличивая T. Integrating по времени, let us find работу, совершенную sourceом.

$$A = \frac{\Phi_0 I_c}{2\pi}\int \sin\varphi\dot{\varphi}dt = \frac{\Phi_0 I_c}{2\pi}\int\limits_0^{\varphi} \sin\varphi d\varphi = \frac{\Phi_0 I_c}{2\pi}(1-\cos\varphi).$$

This work is equal to the energy of the Josephson junction.

Answer: \textbfAnswer:

$$E = \frac{\Phi_0 I_c}{2\pi}(1-\cos\varphi)$$

A10 — 0.20 pts

For a typical Josephson junction, the value of the critical current is $I_c = 0.5$ mA, the resistance is $R = 0.7$ □. Fundamental constants $e = 1.602 \times 10^{-19}$ C, $\hbar = 1.055 \times 10^{-34}$ J ⋅ . Find the numerical value of the energy of the Josephson junction $E(\pi)$ and the frequency of voltage oscillations at constant current through the contact $I = 2I_c$.

$$E(\pi) = \frac{\hbar I_c}{e} = 3.3 \times 10^{-19}\ \text{J},$$

$$\omega = 2\sqrt{3}\frac{eRI_c}{\hbar} = 1.84 \times 10^{12}\ \text{s}^{-1}$$

B1 — 0.30 pts

Let a direct current I flow into the contact (see figure in the introduction to the problem). Write down a differential equation describing the change in phase difference φ. The answer may also include Φ_0, R, C, I_c.

To the expression for the current from part A2, you need to add the current that goes to charge the capacitor, $I_C = \dot{q} = C\dot{V}$, and the voltage derivative is expressed through the second derivative of the phase

Answer: \textbfAnswer:

$$I = I_c \sin\varphi + \frac{\Phi_0}{2\pi R}\dot{\varphi} + \frac{\Phi_0 C}{2\pi}\ddot{\varphi}$$

B2 — 0.60 pts

Let the external current be zero and the phase difference $\varphi \ll 1$. In this case, φ (and therefore other quantities characterizing the contact) will perform damped oscillations. Determine the vibration frequency ω_p.

For small phase values, the equation has the form (after dividing by the coefficient before the second derivative)

$$\ddot{\varphi} + \frac{1}{RC}\dot{\varphi} + \frac{2\pi I_c}{\Phi_0 C}\varphi = 0.$$

Это equation имеет стандартный вид уравнения затухающих колебаний

$$\ddot{\varphi} + 2\gamma\dot{\varphi} + \omega_0^2\varphi = 0,$$

с coefficientами

$$\omega_0^2 = \frac{2\pi I_c}{\Phi_0 C}, \quad \gamma = \frac{1}{2RC}.$$

Then frequency $\omega = \sqrt{\omega_0^2 - \gamma^2}$, Этот answer можно norayandтъ, подставит в equation solution вида $\varphi = Ae^{i\omega t}$, найдя соответствующие значения ω and taking the real part.

Answer: \textbfAnswer:

$$\omega_p = \sqrt{\frac{2\pi I_c}{\Phi_0 C} - \frac{1}{4R^2C^2}}$$

B3 — 0.60 pts

Determine the quality factor of the oscillatory system from the previous paragraph Q. At what values of capacitance in the system are oscillations possible rather than damped motion?

Quality factor can be expressed as

$$Q = \frac{\omega_0}{2\gamma} = R\sqrt{\frac{2eI_cC}{\hbar}}.$$

Колебания возможны, пока value ω_p вещественно, иначе solution уравнения на φ будет представлять собой сумму затухающих экспонент, hence we find

$$C \geq \frac{\Phi_0}{8\pi R^2 I_c}.$$

Answer: \textbfAnswer:

$$Q = R\sqrt{\frac{2eI_cC}{\hbar}}, \quad C \geq \frac{\Phi_0}{8\pi R^2 I_c}.$$

C1 — 1.20 pts

Let the critical currents through the contacts be I_{ac} and I_{bc} respectively. Then the maximum superconducting current I that can flow through such a contact will depend on the magnetic flux through the interferometer. Determine this maximum value I_0, as well as the cosines of the phase differences φ_a, φ_b at which this maximum is achieved. Express your answer using I_{ac}, I_{bc}, Φ, Φ_0. Your answers should not contain inverse trigonometric functions.

The total current through the contacts is

$$I = I_a + I_b = I_{ac}\sin\varphi_a + I_{bc}\sin\varphi_b.$$

We use coratio между фазами $\varphi_b = \varphi_a - \Delta\varphi$, where $\Delta\varphi = 2\pi\Phi/\Phi_0$, we get после преобсований

$$I = I_{ac}\sin\varphi_a + I_{bc}\sin(\varphi_a - \Delta\varphi) = (I_{ac} + I_{bc}\cos\Delta\varphi)\sin\varphi_a - I_{bc}\sin\Delta\varphi\cos\varphi_a = I_0\sin(\varphi_a - \psi),$$

where amplitude

$$I_0^2 = (I_{ac} + I_{bc}\cos\Delta\varphi)^2 + (I_{bc}\sin\Delta\varphi)^2 = I_{ac}^2 + I_{bc}^2 + 2I_{ac}I_{bc}\cos\Delta\varphi,$$

а фаза olimitяется соотношениями

$$I_0\cos\psi = I_{ac} + I_{bc}\cos\Delta\varphi, \quad I_0\sin\psi = I_{bc}\sin\Delta\varphi.$$

Ток будет максимальным, if аргумент синуса equals $\pi/2$, that is $\varphi_a = \psi + \pi/2$, hence

$$\cos\varphi_a = -\sin\psi = -\frac{I_{bc}\sin\Delta\varphi}{\sqrt{I_{ac}^2 + I_{bc}^2 + 2I_{ac}I_{bc}\cos\Delta\varphi}}.$$

Косинус второй фазы можно погayandть, using симметрию между контактами a and b (if записать expression for I_0 via/through φ_b, expression будет отличаться только заменой a на b в индексах and изменением signa $\Delta\varphi$). Иначе его можно найти прямым вычислением

$$\cos\varphi_b = \cos(\varphi_a - \Delta\varphi) = \cos\varphi_a\cos\Delta\varphi + \sin\varphi_a\sin\Delta\varphi.$$

Substituting выражения for косинуса and синуса via/through ψ, we get

$$\cos\varphi_b = -\sin\psi\cos\Delta\varphi + \cos\psi\sin\Delta\varphi = -\frac{I_{bc}\sin\Delta\varphi}{I_0}\cos\Delta\varphi + \frac{I_{ac} + I_{bc}\cos\Delta\varphi}{I_0}\sin\Delta\varphi,$$

$$\cos\varphi_b = \frac{I_{ac}\sin\Delta\varphi}{\sqrt{I_{ac}^2 + I_{bc}^2 + 2I_{ac}I_{bc}\cos\Delta\varphi}}.$$

Answer: \textbfAnswer:

$$I_0 = \sqrt{I_{ac}^2 + I_{bc}^2 + 2I_{ac}I_{bc}\cos\left(2\pi\frac{\Phi}{\Phi_0}\right)}$$

$$\cos\varphi_a = -\frac{I_{bc}\sin\left(2\pi\frac{\Phi}{\Phi_0}\right)}{\sqrt{I_{ac}^2 + I_{bc}^2 + 2I_{ac}I_{bc}\cos\left(2\pi\frac{\Phi}{\Phi_0}\right)}}$$

$$\cos\varphi_b = \frac{I_{ac}\sin\left(2\pi\frac{\Phi}{\Phi_0}\right)}{\sqrt{I_{ac}^2 + I_{bc}^2 + 2I_{ac}I_{bc}\cos\left(2\pi\frac{\Phi}{\Phi_0}\right)}}$$